Hashim Hameed

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WORK EXPERIENCE

Robotics Engineering Intern | Los Alamos National Laboratory *Los Alamos, NM | June 2026 – August 2026*

- Directed team of 4 in development of custom humanoid by defining observation and action spaces, meeting hardware and control constraints of linearly-actuated system
- Developed reinforcement learning pipeline using standalone Nvidia Newton engine, enabling GPU-accelerated parallelization of 4096 environments and computational load reduction of 15% in contrast to Issac Lab
- Constructed weighted reward function for bipedal locomotion of humanoid actor resulting in 98% stable gait

Researcher | UNLV Drones and Autonomous Systems Lab *Las Vegas, NV* | *Nov 2023 – Present*

- Performed MuJoCo based simulation-to-real transfer on 12 DoF custom humanoid with custom trained model based policy, producing 86% stable locomotion given variable command velocity
- Created autonomous vision based detection and manipulation scripting for two 20-DoF humanoid robots, increasing locomotive manipulation stability by 32% compared to single actor
- Utilized ROS motion capture data for 2.4 GHz data transmission to mobile robot platform for movement toward target, improving relative angle calculation by 1.13 factor compared to vision based alternative
- Built microcontroller based game console controller emulator for autonomous mobile robot operation by recreating communication protocol, reducing latency by 6.3% compared to wireless alternative

EDUCATION

University of Nevada Las Vegas **Dec 2026**
Bachelor of Science, Electrical Engineering

PUBLICATIONS

[1] H. Hameed and P. Oh, "Collaborative Rigid Body Manipulation Toward Heterogeneous Humanoids," 2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2026

PROJECTS

Spotlight Cable-Driven-Parallel-Robot (CDPR)

- Developed inverse-kinematic based movement using 4 fixed-position actuator encoders' relation to cable length and end-effector position, achieving 10mm tolerance
- Modeled underactuated open-loop control system in MATLAB, enabling predictive reduction of end-effector orientation drift by 20 degrees on roll and pitch axes
- Designed and fabricated passive gimbal end-effector using Solidworks and additive manufacturing, lowering actuator current draw below 1.5A threshold via 40g weight reduction

Embedded System Development

- Generated compare-timer based PWM signal for DC motor control, with custom GUI permitting user to input speed of 30-130 RPM via USART message and plot both input and encoder based speed values over time
- Produced ADC and temperature monitoring application with 0.5s latency via BLE and 2.4 GHz communication

Electronic Design & Development

- Drafted and built audio effect circuit, manipulating AC waveform through clipping following sequential unity gain of 26.5dB and achieving distorted output with variable characteristics therein

TECHNICAL SKILLS

Languages - C++, C#, Python, Java
Software - LTspice, KiCAD, MATLAB, Quartus II, Fusion 360
Frameworks - Robot Operating System (ROS), Issac Gym, Nvidia Newton, MuJoCo
Hardware - Jetson SBCs, Arduino, Raspberry Pi, TI MSP432 & CC135Rx